EXERCISE 16

Phase line • Stability • Explicit solution

Pitchfork Phase Line and Stability Regimes

Original statement · ProblemsCh2.pdf, p. 4

PROBLEM STATEMENT

Consider the autonomous equation

$$\frac{dx}{dt} = x\left(\kappa^2 - x^2\right)$$

with a general initial condition $x(0) = x_0$.

(i) Calculate the stationary solutions, which satisfy $dx/dt = 0$;
(ii) Calculate and sketch the solutions for $t > 0$, discussing the different behaviors which can be obtained in terms of the relative values of $\kappa > 0$ and x_0, with the help of the uniqueness theorem.

Source: ProblemsCh2.pdf, Exercise 16 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Autonomous equation $\dot{x} = f(x)$ with $f(x) = x(\kappa^2 - x^2) = \kappa^2 x - x^3$ and parameter $\kappa > 0$. We have

$$f'(x) = \kappa^2 - 3x^2.$$

Both f and f' are polynomials, hence continuous on $\mathbb{R}$: the IVP has a unique solution for every $x_0 \in \mathbb{R}$, on a maximal interval of existence. The right-hand side is odd, $f(-x) = -f(x)$, so if $x(t)$ is a solution then $-x(t)$ is also a solution (symmetry $x \to -x$).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why these methods: (a) the phase line (sign of f between equilibria) and the no-crossing consequence of uniqueness give the qualitative picture for every x_0 without solving; (b) the equation is a Bernoulli equation ($\alpha = 3$, see Problem 2.8), so the substitution $w = x^{-2}$ turns it into a linear equation and gives the exact solution to confirm the qualitative picture and to find timescales.

Change of variables written explicitly: $w = x^{1-\alpha} = x^{-2}$ (valid for $x \neq 0$), $w' = -2x^{-3}x'$, $w(0) = 1/x_0^2$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): stationary solutions

Solve $f(x^*) = x^*(\kappa - x^*)(\kappa + x^*) = 0$ using $\kappa^2 - x^2 = (\kappa - x)(\kappa + x)$:

$$x^* = 0, \qquad x^* = \kappa, \qquad x^* = -\kappa.$$

Stability from $f'(x^*)$: $f'(0) = \kappa^2 > 0$ (unstable); $f'(\pm\kappa) = \kappa^2 - 3\kappa^2 = -2\kappa^2 < 0$ (asymptotically stable). Therefore the constant solutions are $x \equiv 0$ (unstable) and $x \equiv \pm\kappa$ (stable).

Part (ii), step 1: phase line from uniqueness

By uniqueness solutions cannot cross the equilibria, so the equilibria $-\kappa, 0, \kappa$ split $\mathbb{R}$ into four invariant intervals. Sign of $f = x(\kappa - x)(\kappa + x)$:

Region	**Signs of x, $\kappa - x$, $\kappa + x$**	**Sign of f**	**Motion**
$x > \kappa$	$+,\ -,\ +$	$-$	decreases to κ
$0 < x < \kappa$	$+,\ +,\ +$	$+$	increases to κ
$-\kappa < x < 0$	$-,\ +,\ +$	$-$	decreases to $-\kappa$
$x < -\kappa$	$-,\ +,\ -$	$+$	increases to $-\kappa$

In each region the solution is monotone and bounded by the neighboring equilibria, hence global in time, with a limit that is an equilibrium: all solutions with $x_0 > 0$ tend to κ, all with $x_0 < 0$ tend to $-\kappa$, and $x_0 = 0$ stays at 0.

Part (ii), step 2: exact solution

Let $x_0 \neq 0$ (then $x(t) \neq 0$ for all t, by uniqueness). Compute $w = x^{-2}$ using the chain rule:

$$w' = -2x^{-3}x' = -2x^{-3}\left(\kappa^2 x - x^3\right) = -2\kappa^2 x^{-2} + 2 = -2\kappa^2 w + 2.$$

Linear equation $w' + 2\kappa^2 w = 2$. Integrating factor: $\int 2\kappa^2\,dt = 2\kappa^2 t$, $\mu = e^{2\kappa^2 t}$, $\mu(0) = 1$, and $(\mu w)' = 2e^{2\kappa^2 t}$. Barrow from 0 to t:

$$w(t)\,e^{2\kappa^2 t} - w(0) = \int_0^t 2e^{2\kappa^2 s}\,ds = \frac{1}{\kappa^2}\left[e^{2\kappa^2 s}\right]_0^t = \frac{1}{\kappa^2}\left(e^{2\kappa^2 t} - 1\right).$$

Hence, with $w(0) = 1/x_0^2$,

$$w(t) = \frac{1}{\kappa^2} + \left(\frac{1}{x_0^2} - \frac{1}{\kappa^2}\right) e^{-2\kappa^2 t} = \frac{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}{\kappa^2 x_0^2}.$$

Since $x = \pm w^{-1/2}$ with the sign fixed by x_0 (a solution cannot change sign), and $x(0) = x_0$:

$$x(t) = \frac{\kappa\, x_0}{\sqrt{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}}, \qquad x_0 \neq 0.$$

(For $x_0 = 0$ the solution is $x \equiv 0$, which the formula does not cover.) Check: at $t = 0$, the square root is $\sqrt{\kappa^2} = \kappa$, so $x(0) = x_0$.

Part (ii), step 3: behavior for $t > 0$

Write $E(t) = e^{-2\kappa^2 t} \in (0, 1]$ for $t \geq 0$. The radicand is $x_0^2(1 - E) + \kappa^2 E$, a convex combination of $x_0^2 > 0$ and $\kappa^2 > 0$, hence positive for all $t \geq 0$: the solution exists for all $t > 0$. As $t \to \infty$, $E \to 0$, the radicand $\to x_0^2$ and

$$x(t) \to \frac{\kappa\, x_0}{|x_0|} = \kappa\ \mathrm{sgn}(x_0).$$

Initial condition	**Behavior for** $t > 0$	**Limit as** $t \to \infty$
$x_0 > \kappa$	decreasing, stays above κ	κ
$x_0 = \kappa$	constant $x \equiv \kappa$	κ
$0 < x_0 < \kappa$	increasing, stays below κ	κ
$x_0 = 0$	constant $x \equiv 0$	0 (unstable)
$-\kappa < x_0 < 0$	decreasing, stays above $-\kappa$	$-\kappa$
$x_0 = -\kappa$	constant $x \equiv -\kappa$	$-\kappa$
$x_0 < -\kappa$	increasing, stays below $-\kappa$	$-\kappa$

Shape of the sketch: for $0 < x_0 < \kappa$ the curve is sigmoidal if $x_0 < \kappa/\sqrt{3}$ (inflection point where $x'' = f'(x)\,x' = 0$, i.e. $f'(x) = \kappa^2 - 3x^2 = 0$, at $x = \kappa/\sqrt{3}$, where the slope is maximal), and concave without inflection if $x_0 \geq \kappa/\sqrt{3}$. The curves for $x_0 < 0$ are mirror images ($x \to -x$).

Behavior backward in time (for completeness): the radicand vanishes when $E = x_0^2/(x_0^2 - \kappa^2)$, which requires $|x_0| > \kappa$ and happens at $t = -\frac{1}{2\kappa^2} \ln \frac{x_0^2}{x_0^2 - \kappa^2} < 0$. So solutions with $|x_0| > \kappa$ blow up in finite negative time, while those with $0 < |x_0| < \kappa$ exist for all $t < 0$ and tend to 0 as $t \to -\infty$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the ODE by differentiating the explicit solution. With $D(t) = x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}$: $x = \kappa x_0 D^{-1/2}$, $x' = -\frac{1}{2}\kappa x_0 D^{-3/2} D'$, and $D' = -2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. Also $\kappa^2 - x^2 = \kappa^2 - \frac{\kappa^2 x_0^2}{D} = \frac{\kappa^2(D - x_0^2)}{D} = \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$, so $x(\kappa^2 - x^2) = \kappa x_0 D^{-1/2} \cdot \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$. This equals x': $-\frac{1}{2}\kappa x_0 D^{-3/2} \cdot \left(-2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}\right) = \kappa x_0 D^{-3/2}\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. They agree.

Numerical check (computed): for $\kappa = 1$, a fourth-order Runge-Kutta integration over $t \in [0, 5]$ (20000 steps) matches the closed form to better than 10^{-14} for $x_0 = 0.3,\ 2,\ -0.5,\ -3$. At $t = 5$ the distance to $\kappa = 1$ is 2.3×10^{-4} for $x_0 = 0.3$ and 1.7×10^{-5} for $x_0 = 2$, consistent with the time constant $1/(2\kappa^2) = 0.5$.

Time scale and dimensions: the linearization at $x = \pm\kappa$ has rate $f'(\pm\kappa) = -2\kappa^2$, so the approach to $\pm\kappa$ is exponential with time constant $1/(2\kappa^2)$ (visible in the factor $e^{-2\kappa^2 t}$), and at $x = 0$ the repulsion rate is κ^2. The equation is written in non-dimensional form: κ^2 is a rate with units $[t]^{-1}$, and x is measured in units for which the cubic coefficient equals 1.

Bifurcation context (Robinson, Section 7.6): for the parameter $\mu = \kappa^2$ the equation $\dot{x} = \mu x - x^3$ has one equilibrium $x = 0$ (stable) for $\mu < 0$, and for $\mu > 0$ the origin becomes unstable and two stable branches $\pm\sqrt{\mu}$ appear: a supercritical pitchfork. Here $\kappa > 0$ is the case $\mu > 0$.

RESULT

Equilibria 0 (unstable) and $\pm\kappa$ (stable). For $x_0 \neq 0$: $x(t) = \kappa x_0 / \sqrt{x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}} \to \kappa\ \mathrm{sgn}(x_0)$ as $t \to \infty$.